Fiscal Year:	FY 2026	**Task Last Updated:**	FY 01/30/2026
PI Name:	Cornell, Eric Ph.D.		
Project Title:	Zero-G Studies of Few-Body and Many-Body Physics		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80309-0440	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013 Fundamental Physics NNH13ZTT002N (Cold Atom Laboratory--CAL)
Start Date:	04/01/2014	**End Date:**	09/30/2028
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JPL
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 9/30/2028 per J. Callas/JPL (Ed., 6/24/25). NOTE: End date changed to 9/30/2026 per D. Griffin/NASA-HQ (Ed., 1/20/25). NOTE: New contract number is JPL 1700833 per J. Callas/JPL (8/29/24). NOTE: End date changed to 9/27/2025 per J. Williams/JPL (Ed., 7/22/24) NOTE: End date changed to 9/27/2024 per U. Israelsson/JPL (Ed., 10/20/21) NOTE: End date changed to 8/31/2021 per U. Israelsson/JPL (Ed., 5/12/2020) NOTE: End date changed to 4/30/2020 per PI (Ed., 5/1/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Engels, Peter Ph.D. (Washington State University, Pullman) Mossman, Maren Elizabeth Ph.D. (University of San Diego) D'Incao, Jose Ph.D. (University of Colorado Boulder)		
Grant/Contract No.:	JPL 1700833 ; JPL 1502690		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	Future advances in both technology and fundamental science will hinge on a better understanding of the weird effects of quantum mechanics on collections of electrons, atoms, molecules, and so on. In some cases, experiments probing this so-called “quantum few-body and many-body physics” can be more readily accomplished in the weightless environment found in an orbiting laboratory. We propose a staged series of experiments, including (1) “first science” experiment, to be performed in the Cold Atom Laboratory (CAL) flying in the International Space Station (ISS) first-generation, to answer a question in few-body quantum physics that can’t be performed in a ground-based laboratory: how universal are the weakly bound clusters of three atoms known as Efimov trimers? In a weightless environment, experiments can be performed at very low densities and temperatures, the perfect conditions for these exotic but fragile quantum states to form. (2) Bose gases with “infinite” interactions. As interactions between atoms become stronger, there is a crossover between gas-phase and liquid behavior. In ultra-cold atoms, the crossover is between a quantum liquid and a quantum gas. (3) Highly rotating quantum gases. Many of the most exotic and unexplored predicted states of matter occur in the presence of very strong magnetic fields, for electrons, or high rates of rotation, for neutral particles. We will explore Quantum Hall physics in highly rotating Bose and Fermi gases. Experiments (2) and (3) will benefit significantly from the longer expansion times and weaker traps possible in weightlessness. Preliminary versions of both experiments will be done in a ground-based laboratory in order to establish the foundation for future flight-based experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Physics is the discipline that provides understanding of biology and chemistry at the most microscopic level, and the area within physics most relevant to chemistry and biology is “few-body physics.” It is an often neglected portion of physics, because it is so difficult to do! An important way to make progress is to simplify, simplify, simplify: to come up with model systems in which we can make progress that can later be applied to human-centric disciplines like biology, and develop exotic and useful new materials. A promising way to simplify is to study matter at lower temperature, and lower densities. The Cold-Atom Lab (CAL) flying in the International Space Station (ISS) is where we will reach the lowest possible temperatures, and low densities, to do our studies of simple, yet intricate (think “snowflakes”) clusters of three or four atoms. We have been doing prefatory experiments and calculations here on Earth. Not at as low temperature, but still cold enough to help us learn things we will need to know to do the space experiments. While CAL is now in flight, we have been participating in the effort to remotely tune it up for maximum performance and are progressing with few-body experiments using CAL.
	The Cold Atom Laboratory (CAL) apparatus is a unique platform designed to conduct sophisticated experiments with laser-cooled atoms aboard the International Space Station (ISS). The CAL hardware includes several delicate components, such as an ultrahigh vacuum chamber to confine and shield the atoms, a complex laser system, control electronics, and magnetic field coils. The system is entirely remote-controlled from the Jet Propulsion Laboratory (JPL), marking a significant milestone in quantum technology and heralding the era of microgravity-based quantum experiments. A microgravity environment offers fascinating new perspectives for the investigation of intricate quantum mechanical few-body physics. The study of new phases of matter as well as applications to interferometric measurements are two examples for highly promising paths forward with direct relevance for topics suggested by the recent Decadal Survey. A key element of CAL is the atom chip, which is patterned with traces that carry electrical currents. These currents generate magnetic fields that confine the ultracold atoms. Together with the surrounding vacuum chamber, this forms a science module. As CAL experiments progress, the science modules can be upgraded to enhance the capabilities of the apparatus. In 2024, a new science module, SM1, was installed on the ISS following a previous module, SM3b, that had a vacuum problem compromising its performance. Throughout 2024 and 2025, our team worked closely together with the CAL scientists and engineers to evaluate the performance of the SM1 module and assess its suitability for our potassium experiments. While an initial Magneto-Optical Trap (MOT) of potassium atoms was achieved using SM1, electronic glitches with current driver modules and the computer control needed to be identified and mitigated in 2025. While SM1 successfully produced rubidium Bose–Einstein condensates, its operation with potassium was hindered by these instabilities of the apparatus. As a successor to SM1, a new module, SM3X, was developed on the ground to overcome the technical limitations of SM1 and to enable efficient potassium research on CAL. During the design and construction of SM3X, our team remained in continuous communication with JPL scientists and engineers to ensure that the specifications of SM3X would meet our experimental requirements. These discussions covered numerous technical details, including current-driver connection diagrams, the placement and frequency range of the microwave antenna, and other design considerations. Our team has performed detailed calculations confirming that the new system’s capabilities will be sufficient. The CAL apparatus confines atoms in close proximity to an atom chip patterned with current-carrying electrical traces. These currents generate magnetic fields that, together with additional fields produced by larger external coils, create the trapping potential for the ultracold atoms. A critical ingredient in evaluating the performance of the system is the ability to numerically model the detailed geometry of both the atom-chip conductors and the surrounding magnetic field coils. Such simulations are essential for determining the precise trapping strengths and field gradients, which influence every stage of the experimental sequence. Our team has developed a custom suite of numerical tools that compute these fields with high accuracy. A distinguishing feature of our approach is that all conductors—including the wiring of the external bias coils—are treated exactly, without simplifying approximations. Because each science module (such as SM1 and SM3X) features a distinct layout of electrical traces, we continually update our model to reflect these differences. An additional numerical challenge arises from manufacturing tolerances, particularly in the exact positioning of the atom chip relative to the external coils. By simulating atomic trajectories within the trap, we can infer the true geometric parameters of the apparatus—details that are otherwise difficult to measure directly. Modeling various aspects of the CAL apparatus has been an ongoing collaborative effort among our team and the other PI groups involved in the project. In 2024, the collaboration initiated a shared document to consolidate SM1-related technical information, including detailed chip-coordinate data. Work on this common resource continued through 2025, incorporating additional discussion on topics such as the resolution of the wide-field camera and the basic programming of different atom-trap configurations. We remain in active communication with the other CAL PIs to evaluate various trapping schemes and to benchmark our numerical models against experimental measurements.

Task Progress:	An intriguing feature of CAL is that, despite being a highly complex quantum-science instrument, it is operated entirely by remote control from Earth. In standard operation, each experimental run, lasting roughly one minute, is executed using a newly uploaded program that specifies the full sequence of actions a Principal Investigator (PI) team wishes to perform. Efficiently generating these programs is therefore essential. Each run involves a large number of coordinated steps, such as opening and closing laser shutters, adjusting electrical currents, triggering imaging systems, and applying radiofrequency or microwave pulses to the atoms. To streamline this process, our team has developed and continually refined a software suite that automates many aspects of program generation. As CAL evolves through successive science modules (e.g., from SM3b to SM1 and, soon, to SM3X), we update and extend this tool to accommodate new hardware capabilities and requirements. We have also conducted theoretical studies, led by Prof. Jose D'Incao at the University of Massachusetts, Boston (UMB), to better understand finite temperature effects affecting the dynamics in ultracold quantum gases—an important aspect of our planned experimental research with CAL. Being able to theoretically describe these effects is crucial to connect with future experimental data and to provide supporting evidence for the universality of such observations. In this context, universality means that systems can exhibit the same observable behavior even if they are governed by very different underlying microscopic physics. We have developed a novel, highly efficient calculational approach for this. A publication is in preparation. This work has been developed by theory postdoc Michael D. Higgins at UMB, who has been partially funded by this program. Besides these developments, our theoretical studies have also provided a deeper understanding of spin effects and have investigated universality in three-body systems from a broader perspective. In collaboration with Johannes Denschlag's group at Ulm University, we have developed a better understanding of relevant molecular dynamics influencing our planned experiments, providing new forms of control over the system. We have demonstrated this scheme both theoretically and experimentally, thus confirming the accuracy of our newly developed theoretical framework for describing three-body phenomena. The extension of the concept of universality for ionic three-body systems has provided us with a broader view on universal phenomena and has allowed us to identify potential applications. Overall, our team has contributed to CAL on multiple levels. We have participated directly in the flight-based research using the CAL apparatus aboard the ISS, while also conducting complementary ground-based experiments that serve as fiducial references for future CAL measurements in new parameter regimes. Our theoretical efforts have helped identify the most promising scientific directions for the potassium work, and develop the analytical tools required for interpreting forthcoming data. This work has been carried out collaboratively across four institutions: JILA/University of Colorado (Eric Cornell), Washington State University (Peter Engels), the University of San Diego (Maren Mossman), and the University of Massachusetts Boston (José D'Incao). The project has also provided an exceptional platform for student education and training. In 2025, graduate students Colby Schimelfenig, Jiahao Qiu, and undergraduate researcher Jake Pierson were actively involved in the CAL-related work at Washington State University. At the University of San Diego, Professor Mossman collaborated with undergraduate student Mark Gin (now pursuing graduate studies at USC) to plan experiments and gain experience operating the apparatus remotely. Students regularly participated in online meetings with the JPL team and contributed to technical conversations within their local PI groups. At the University of Massachusetts, Boston, postdoctoral researcher Michael D. Higgins was partially supported by this program to conduct theoretical studies supporting the CAL experiments. CAL-related material has also been incorporated into several of our outreach events aimed at engaging younger students in science. A formal presentation of our recent work was delivered by Peter Engels at the 41st Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), held in Phoenix, AZ, in December 2025. Jose D'Incao presented theory results at several events, including the ITAMP/Harvard Workshop, "Fundamental three-body processes in physics and chemistry" (ITAMP/Harvard, June 2025), and the "Quantum Collaborations" event (University of Massachusetts Boston, April 2025). [Ed. Note: See Bibliography.]

Bibliography Type:	Description: (Last Updated: 04/15/2026)
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Abstracts for Journals and Proceedings	D'Incao JP. "Quantum-state resolved chemical reactions in ultracold quantum gases." "Fundamental three-body processes in physics and chemistry", a workshop sponsored by the Institute for Theoretical Atomic, Molecular and Optical Physics (ITAMP) at Harvard University, Cambridge, MA, June 2-4, 2025. Abstracts. "Fundamental three-body processes in physics and chemistry", a workshop sponsored by the Institute for Theoretical Atomic, Molecular and Optical Physics (ITAMP) at Harvard University, Cambridge, MA, June 2-4, 2025. , Jun-2025
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